

How To Science 101

2 topics: Image processing – AI from scratch

Final deliverables

Challenges of the project

Recommendations

Linearity in a non-linear world

The AI dilemma...

Learning in the AI age

Bonus: How to design complex (and critical) systems



Welcome to ...

How To Science 101 – M1 AM Modeling project

Overview

Duration: **4 weeks**.

Programming language: **C++** (to promote clean code design and structure).

No Python – **EXCEPT** – for **experimentation** with filtering, optimization, ... AND for **plotting** data (graphs, animations of iterations, ...)

0 Python in the final “product”.

Teamwork: **teams of ~4 people**.

Your goal will be to learn - **as a team** - to:

- **Analyze** = investigate a new problem
- **Code** = implement your solutions
- **Iterate** = experiment, evaluate and improve them, and
- **Disseminate** = chose the most significant contributions among your work, put them forward in a "scientific communication package" and share them with the rest of the world!



Topics: Image processing – AI from scratch

Project A – Image processing

Use classical algorithms (Fourier, linear algebra) to solve problems related to fingerprint recognition.

POSSIBLE FUN GOAL → “Ex-boyfriend/girlfriend removal application” (as an extension of the “in-filling” problem)

NB: If you pick project **A** and **FINISH** the required tasks, you are **free to experiment with AI based solutions!**

Project B – AI from scratch

No external libraries! (except the basic ones, stdlib etc.)

POSSIBLE FUN GOAL → Automatic detection of handwritten digits → “Blackboard calculator”

NB: Digits recognition is definitely doable ...
“Blackboard calculator” will require a bit more work
(but still doable!)



Final deliverables

- A **20 to 30 min** oral presentation (everyone must speak), in English.

NB: No need to RECOMPILE your project during the demo...

Structure

- 15 or 20 min slides (TO BE DEFINED)
- 5 ~ 10 min **live demo** of your project (GUI or command line, as long as **computations are carried out in real-time** at demo time).

The oral presentation will be delivered **in front of the entire class**. Attendance compulsory.

- A **7~12 pages scientific article** (single or double column, docx or latex (TO BE DEFINED)) presenting your work (see details below).

Complete structure:

- **Title + authors** + fictitious affiliation
- **Abstract:** summary of the article in max 200 words.
- **Introduction:** general presentation of the context.
- **Motivation:** presentation of ONE (1) specific topic within the introduced context that presents a challenge of interest.
- **Methods:** detailed description of the methods chosen to solve the challenge.
- **Results:** results achieved with the chosen method + performance assessment (computation time, accuracy, code quality, ...).
- **Discussion:** synthesis of the results and analysis of the reasons for success or failure of the proposed methods.
- **Conclusion and future works:** TL;DR / short summary of the discussion and possible paths for improvement.
- The article will be submitted to the fictious "International Journal of M1 AM Projects" (editor: me :-)) and will automatically be accepted for publication.
- Deadline for submission: XX/XX/202X (TO BE DEFINED)
- All articles will be shared with everyone in the group

- An **archive of your project** = source code + all relevant documentation (pptx, docx, pdfs) → zip file or github repo.

- [OPTIONALLY] A **poster** → A helpful exercise for those aiming for a PhD

About the articles

It is rather **unlikely** (but NOT impossible) that in 4 weeks you will discover a novel solution to well known problems. If you do so, I will happily help you ACTUALLY publish your work.

If not, then we will **pretend** that your work is "publication worthy" (for the sake of the exercise).

A few possible angles for a scientific article, along with typical titles:

1) Proposition of technical implementation of a solution to a problem:

*"[Name of your framework]: A Novel **Framework** for Computing ..."*

2) Proposition of theoretical solution to a problem:

*"[Name of your algorithm]: A **Novel Approach** for Solving ..."*

3) Proposition of a literature review of existing solutions to a problem:

*"A **Survey** of Methods for Solving ..."*



Saint Matthew and the Angel – Caravaggio – 1602

Challenges of the project

You will have to overcome the following challenges:

- **Teamwork challenge**: clearly define your roles; choose the best collaboration tools and team interaction mode.
- **Scientific challenge**: understand the solutions in the literature, come up with your own ideas.
- **Mathematical challenge**: understand and solve the mathematical challenges.
- **Implementation challenge**: structure, test, comment, share your code.
- **Communication challenge**: share your knowledge with everybody through an oral presentation and a short scientific paper.

NB: Share your “mix of expertise” so that everyone in the group grows

Math experts → teach code geeks how to math
Code experts → teach math nerds how to code

Project goals

- 1) Experiment the excitement of working, discovering and learning new stuff by yourselves.
- 2) Experience the difficulty of **structuring** and **sharing** your newly acquired knowledge.
 - ➔ **Cycle: expansion / contraction, dispersion / structuration**
- 3) Incidentally, do some **cool coding and math...**
- 4) The project deliberately has a **weak structure** to leave room for you to **experiment**



Recommendations

- **Don't focus on the final grade**: focus on learning. If you **learn new stuff AND contribute to teamwork** → the grade will be good.
- **Don't cheat**: (copy paste from ChatGPT, GitHub, ...) but use the available tools, collaborate and look for inspiration → **You must understand the things you do**.
- **Remember**: you have to “suffer” to learn (science says so: <https://cacm.acm.org/news/the-impact-of-ai-on-computer-science-education/>)

About the final grade: you will be evaluated on what you have actually learned and understood + overall invested energy in the project + **quality (form) of the presentations** (oral + written).

About the quality of your presentation: it is **ESSENTIAL** to learn how to share your knowledge through **compelling oral presentations** and **written** publications.

No matter how smart you are, people will mostly judge you by “your cover”.

Form matters. A lot.

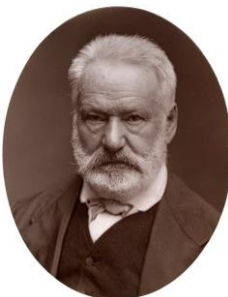
There's **form** = text formatting, typos, grammatical errors... **THIS IS NON-NEGOTIABLE**

But there's ALSO **form** = **structure** of the text, **organization** of the presentation, smooth **transitions**, sound **development** of the reasoning

The **quality of structural form** (organization of ideas) **reflects the quality of your thinking process**.

In other words, TO SOME EXTENT it does make sense to judge a book by its cover because **the cover still says something about the book...**

"La forme c'est le fond qui remonte a la surface" - Victor Hugo

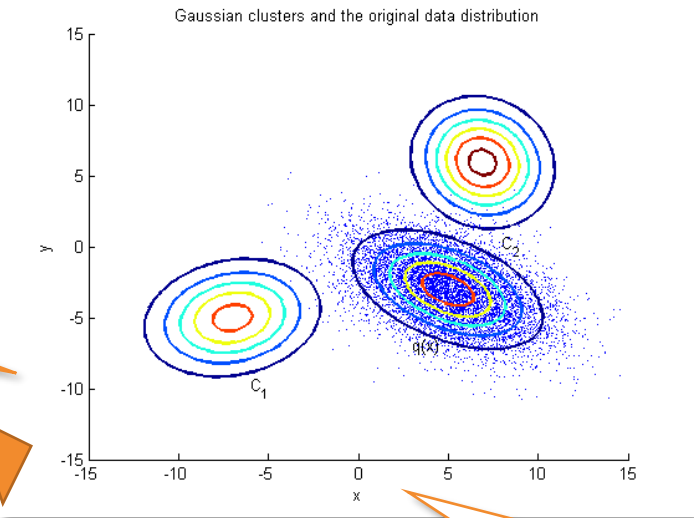


Linearity in a non-linear world

Let's develop an intuition about the **benefits** of AI over **traditional** approaches.

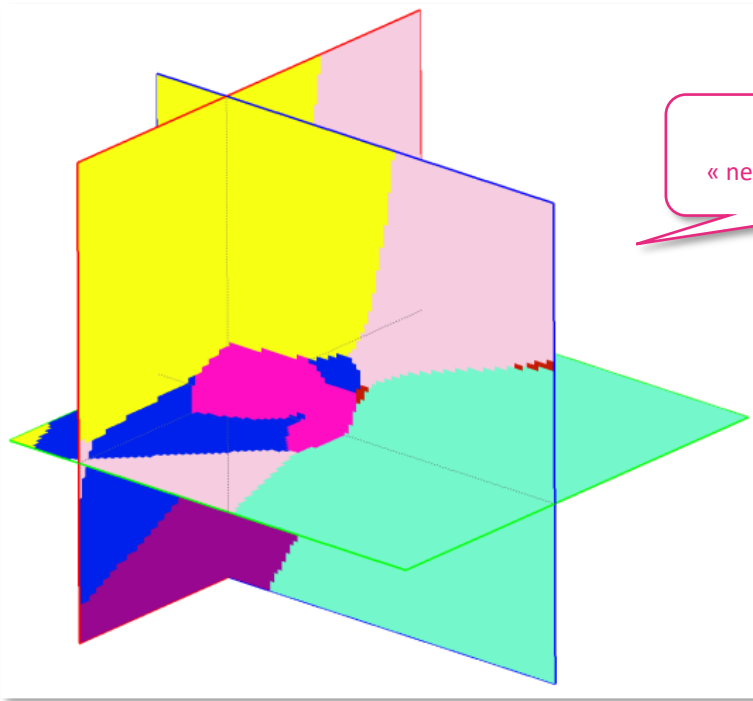
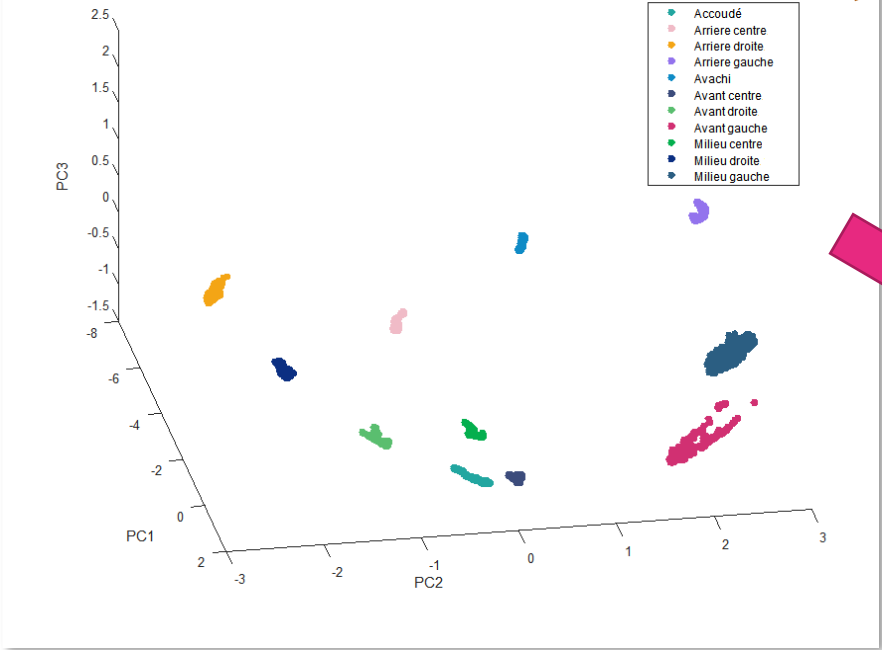
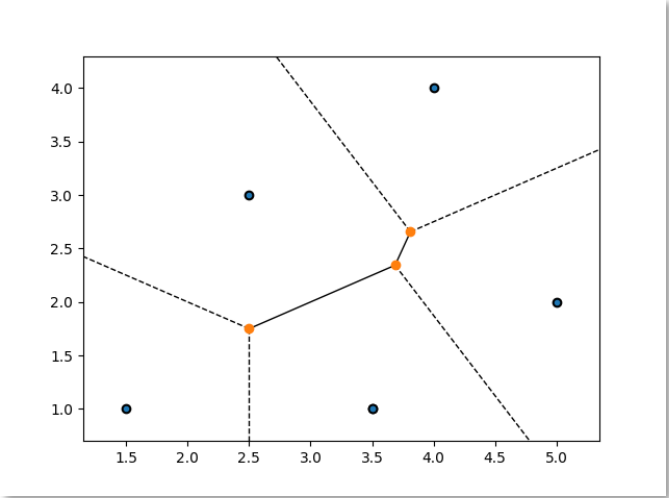


Statistical model
Bayesian classifier
Gaussian mixture



NOT an
explicative model

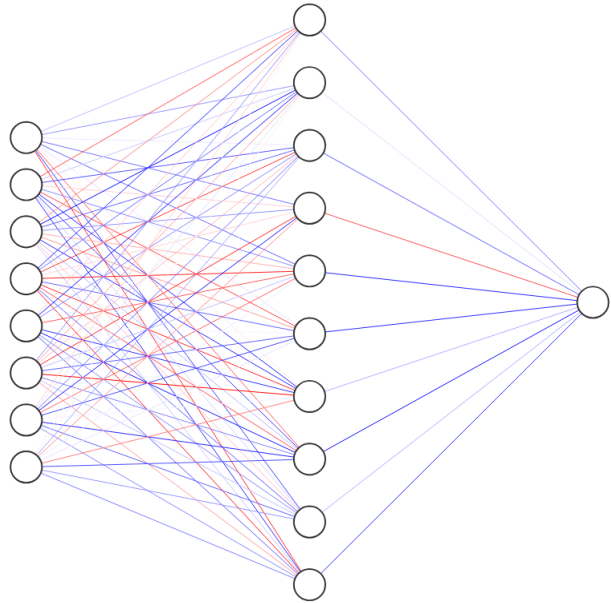
TRADITIONAL → LINEAR CLASSIFIER



Empirical model
« neurons », « activations »

NOT an
explicative model either

AI → NON-LINEAR CLASSIFIER



Input Layer $\in \mathbb{R}^8$

Hidden Layer $\in \mathbb{R}^{10}$

Output Layer $\in \mathbb{R}^1$

<https://alexlenail.me/NN-SVG/>

A quick glance at the sheer size of a typical AI's space parameters

Let's consider a NN with **8 entries** and **10 neurons** ...

→ $8 \times 10 = 80$ **connections**, each bearing a **weight** (this is a **VERY SMALL** neural network)

→ **~ 80 parameters** (weights)

Each weight between **-1 and +1**, sampled by **10 values** (this is **SUPER coarse**... but let's use that for the sake of demonstration).

Total “possible states” = **1e+80**.

Now ...

"In astrophysics, the **Eddington number** is the **number of protons in the observable universe**. Eddington originally calculated it as about 1.57×10^{79} current estimates make it approximately 10^{80} ."

https://en.wikipedia.org/wiki/Eddington_number

So 1e+80 is a lot

ChatGPT has **1.76 trillion parameters** = 1.76×10^{12} = a million millions...

Since I'm a **transformer-based** model, I have many layers of "neurons" in the form of dense units (for feedforward layers) and attention heads (for attention layers). Let me break it down a bit:

Layers and Units in a Transformer Model:

1. **Layers:** GPT-4 has many layers (potentially hundreds), with each layer consisting of several "neurons" or units.
2. **Hidden units per layer:** Each transformer block has a certain number of hidden units (often referred to as the **hidden size**). In large models, this number can range from **thousands to tens of thousands** per layer. For instance, in GPT-3, the hidden size was **12,288 units** per layer.
3. **Number of Attention Heads:** Each layer of the transformer also has multiple attention heads (in GPT-3, there are 96 attention heads in the largest model). Each attention head has its own set of parameters and can be thought of as a specialized "neuron-like" unit that learns how different parts of the input relate to each other.

Estimating Neurons from Parameters:

Given that GPT-4 has **1.76 trillion parameters**, and a large number of these parameters correspond to the connections (weights) between "neurons," the total number of neuron-like units is likely in the **tens of billions** range.

- For comparison, GPT-3 (175 billion parameters) has an estimated **96 layers** and **12,288 hidden units** per layer. This gives you an idea of the order of magnitude, and GPT-4 is significantly larger.

While it's hard to give an exact count of neurons without knowing the specific architecture details of GPT-4, you can roughly think of the network as having **billions to tens of billions of "neurons"** distributed across its layers.

The AI dilemma...

1) The good

Many **actual complex problems** are solved using AI based approaches:

- healthcare (medical image analysis, diagnosis),
- engineering (design exploration),
- numerical modeling (reduced order modeling),
- ...

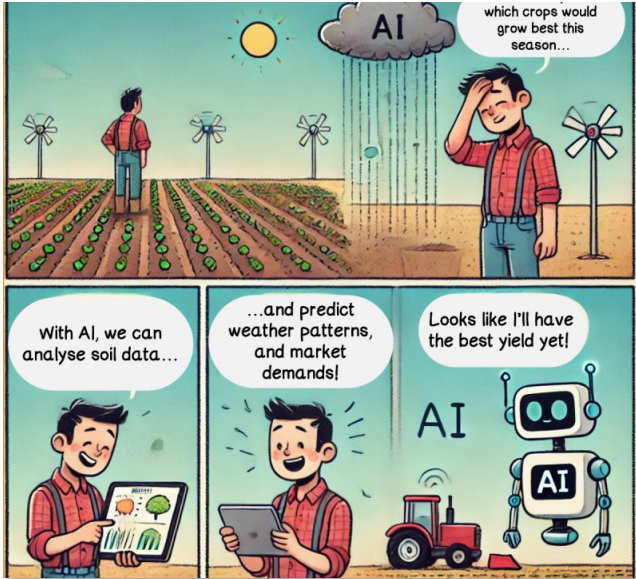
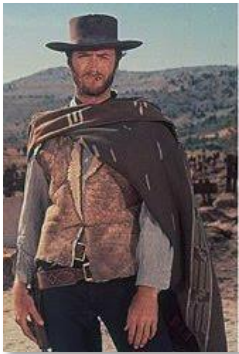
In consumer products: **productivity boosters** such as AI-assisted software development (**DRAWBACKS!**), or ChatGPT to **survey the internet** on **consensual topics**.

AI for good

<https://www.linkedin.com/pulse/use-ai-agriculture-transforming-future-food-christophe-j-bianchi-umkwf>

Entertainment industry

<https://oasis.decart.ai/starting-point?journey=Rugged+Coastline>



The AI dilemma...

2) The bad

For most people (including engineers) AI is a **black box**.

AI is now **universally considered as a "magic wand"** that can solve **EVERY problem**. This diverts attention from classical solutions **with stronger theoretical background**.
<https://www.gartner.com/en/newsroom/press-releases/2024-08-21-gartner-2024-hype-cycle-for-emerging-technologies-highlights-developer-productivity-total-experience-ai-and-security>

AI failures are **hard to correct**; training has to be repeated / fine-tuned.
<https://www.gartner.com/en/articles/what-it-takes-to-make-ai-safe-and-effective>

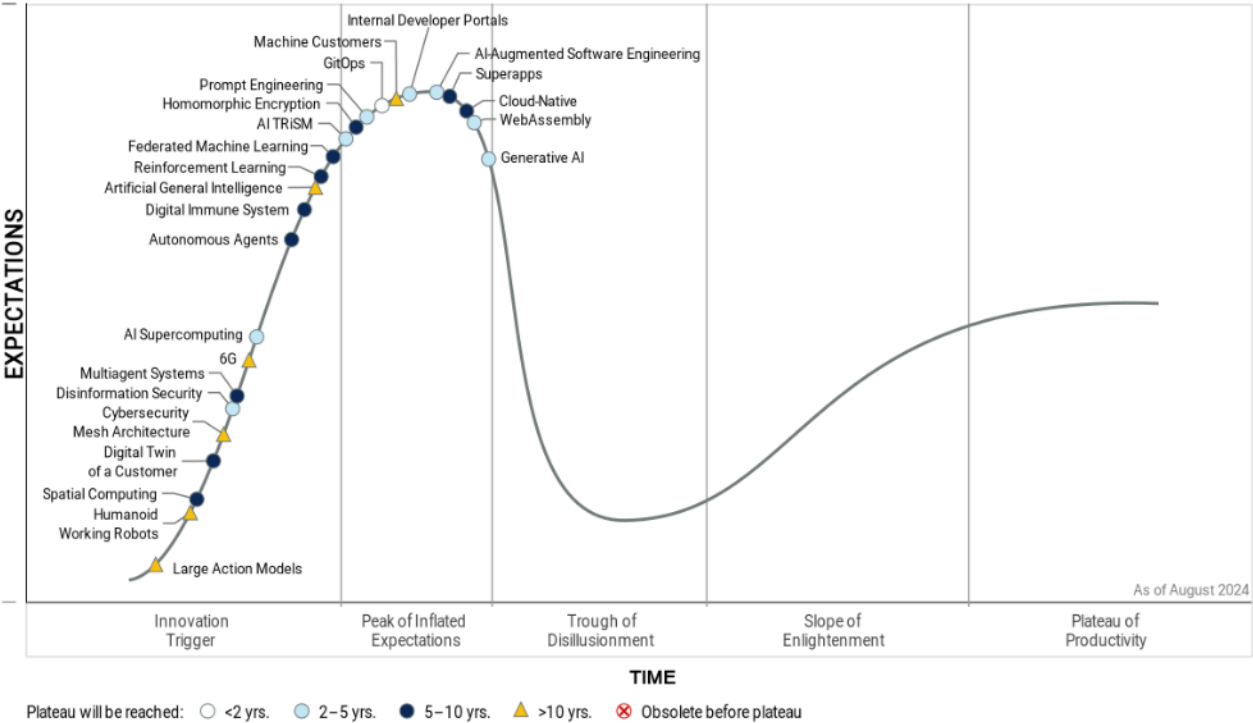
In the industrial world, AI has become a **buzz-word** and this trend has led to a misleading definition of **AI vs Machine Learning**.

In the academic world, AI-based projects are **draining funding**.

This "Anything is AI" state of mind **blurs the lines** between **all possible solutions** to a given problem.

Students are more attracted by the **wonders of the all-powerful AI** algorithms and neglect **fundamental mathematics**, statistics or other traditional approaches.

Oversimplification ... lack of nuance ... lack of originality
<https://substack.com/inbox/post/151222502>



The AI dilemma...



3) The ugly

- **Mass manipulation**: internet bots on social networks, social networks training gen AI on user content, ...
- **Post-truth era**: deep-fakes (Generative AI...)
- **Mass surveillance**: smart cameras during the Paris Olympic games
- **Delegation of high-level competence** to AI algorithms → thinking, writing, structuring ... meaning humans are **losing their ability to perform these cognitive tasks**
- **Energy / resources (water) consumption** will soon be a problem too (if not already)

The technology is **privatized** and **under the control of the capital AND CERTAIN states**. It can become a dystopic **economic** and **political** player.

Human biases are **baked into** the product... So AI is **not a “voice of wisdom”** ... it merely reflects our human opinions.

It is **not sentient** and has **no moral or reasoning**.

Bottom line: For these reasons it is **PARAMOUNT** for **math & IT students** to have a **DEEP UNDERSTANDING** of **how AI works** as you **WILL** play an **active role** in the future of tech.

Learning in the AI age

Why are you here?

- To get a **degree** ... ?
- To **learn** ... ?



Degree = **result** of you learning new stuff

Studies time (University) VS Professional time (Job)

- **Studies time** = **LONG** → Learning new skills takes time (**well-meaning** environment – work on your **weak points**)
- **Professional time** = **SHORT** → Exigence de performance (**competitive** environment – work on your **strong points**)

It takes time, but that's OK

AI is an **accelerator** (... and **facilitator** of good grades!)

However ...

<https://www.radiofrance.fr/franceculture/podcasts/la-conversation-scientifique/qu-est-ce-que-l-ia-nous-fait-4047114>

In French...



➡ You should be **worth more** than a AI user



Learning in the AI age

However ...

- AI is a **productivity tool** widely used in industry
- AI provides access to **personalized explanations** and **tutoring**

AI usage rules during this project

If you **wouldn't** ask it from a **medium/good friend**
don't ask it from **Chat GPT**

Examples



- Could you **write for me** a Quick Sort **in C++** optimized in memory usage?
- Could you **generate for me** a diagram representing the architecture of **my code**?
- Could you **write for me** the outline of **my report** about ... ?

You're not learning **anything**

You're missing the **opportunity** to **VERIFY** your design

You're missing the opportunity to better understand and improve what you did



AI = personal **mentor**

You discover new options

You get some inspiration but the actual work is still on your hands

- Can you **explain** with a **few examples** the **principles** of Quick Sort?
- Can you tell me which **are** the **most commonly** used diagrams to represent software architecture?
- Can you show me a **few examples** of **typical** report outlines about ... ?

Learning in the AI age

Finally ...

« Le danger n'est pas dans la **multiplication des machines** mais dans le **nombre toujours croissant d'hommes habitués dès l'enfance à ne rien vouloir de plus** que ce que les **machines peuvent donner**. »

« Danger is not in the **abundance of machines** but in the **ever-increasing number of people** accustomed **since childhood** (or studies) to **never expect more** than what the **machines can provide**. »

Georges Bernanos (1947)



Brazil, Terry Gilliam, 1985

Bonus: How to design complex (and critical) systems

Example

Optional presentation
~ 45'

